Co-chain complex index going up.
chain complex 下标
co-chain complex 上标

cocycles Z^k coboundaries B^k cohomology $\mathcal{H}^k$

co-chain complex 的常见构造是从一个 chain complex 取对偶

vec. space
$L: V \longrightarrow W$ 线性算子
$L^*: W^* \longrightarrow V^*$
$L^* w^*(v) = w^*(Lv) \quad \forall v \in V, w^* \in W^*$

chain complex $0 \longrightarrow C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_1} C_0 \longrightarrow 0$

cochain complex $0 \longleftarrow C_n^* \xleftarrow{\partial_n^*} C_{n-1}^* \longleftarrow \cdots \longleftarrow C_0^* \longleftarrow 0$

$V^k = C_k^* \qquad \partial^{k-1} = \partial_k^*$

Example: DeRham Complex

1-D: $0 \longrightarrow \underset{V^0}{C^\infty(\Omega)} \xrightarrow{\frac{d}{dx}} \underset{V^1}{C^\infty(\Omega)} \longrightarrow 0$

V^0 即光滑函数空间 Ω 是一维区间 $\frac{d}{dx}$ 是求导算子
显然满足 chain complex 的定义 $\dim \mathcal{H}^0 = 1 = \dim \mathbb{R}/\{0\}$ 说明只有1个连通分支

2-D: $0 \longrightarrow C^\infty(\Omega, \mathbb{R}) \xrightarrow{\text{grad}} C^\infty(\Omega, \mathbb{R}^2) \xrightarrow{\text{rot}} C^\infty(\Omega) \longrightarrow 0$
rot 标量 curl $\quad \text{rot}(u_1, u_2) = -\partial_1 u_2 + \partial_2 u_1 \quad \text{rot}(\text{grad}\, u) = 0 \rightsquigarrow$ complex

3-D: $0 \longrightarrow C^\infty(\Omega) \xrightarrow{\text{grad}} C^\infty(\Omega, \mathbb{R}^3) \xrightarrow{\text{curl}} C^\infty(\Omega, \mathbb{R}^3) \xrightarrow{\text{div}} C^\infty(\Omega) \longrightarrow 0$
$\text{curl}(\text{grad}\, u) = 0 \qquad \text{div}(\text{curl}\, u) = 0 \rightsquigarrow$ complex

n-D: $0 \longrightarrow \underset{\text{zero forms}}{\Lambda^0(\Omega)} \xrightarrow{d} \Lambda^1(\Omega) \xrightarrow{d} \Lambda^2(\Omega) \xrightarrow{d} \cdots \xrightarrow{d} \Lambda^n(\Omega) \longrightarrow 0$
1st exterior derivative

$\Lambda^k(\Omega)$ 是 Ω 上光滑 k 次微分形式构成的 vector space
$d: \Lambda^k \longrightarrow \Lambda^{k+1}$ 满足 $d \circ d = 0 \rightsquigarrow$ cochain complex

如何对单纯形上的微分形式进行积分：需要满足微分形式是 k 次形式则单纯形也要是 k 维的
rmk: 即多元微积分中 $\int f\, dx_1 \wedge dx_2 \wedge \cdots dx_n$ 与被积区域 D 维数一致

故 $\Lambda^0(\Omega)$ 即在点上积分 即0-形式积分就是在点上取值
$\Lambda^1(\Omega)$ 即向量场 可在边上积分 $v \in \Lambda^1(\Omega)$ $f \subset \Omega$ $\dim f = 1$ f 切向量
$\int_f v = \int_a^b v \cdot t\, ds$
$f = [a, b]$
⇓ 类似的
$\Lambda^k(\Omega)$

Stokes theorem $\int_C du = \int_{\partial C} tr\, u$, $u \in \Lambda^{k-1}$, $C \in C_k$

$du \in \Lambda^k$, $\partial C \in C_{k-1}$

外微分如何计算待补充！

外微分的一个重要特点在于适用于任何光滑流形上的求导和积分